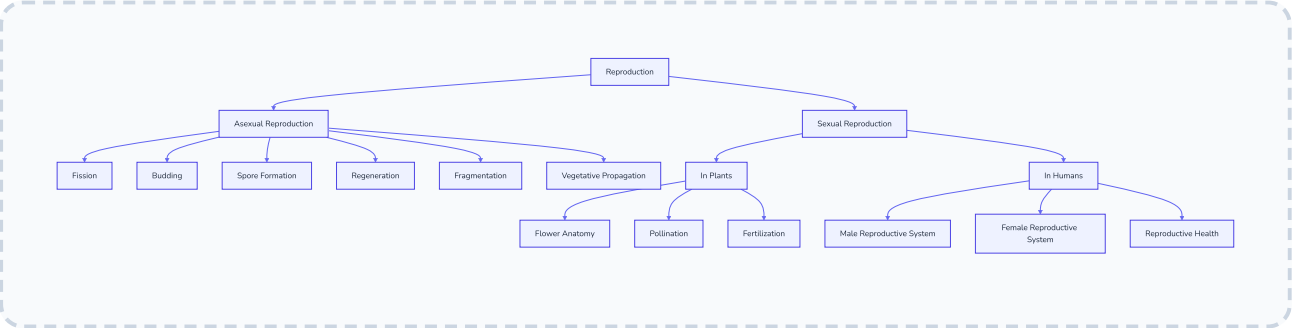
Chapter 7: How do Organisms Reproduce?

🌱 👼

Reproduction is the biological process by which existing organisms give rise to new organisms of the same species. It is essential for the survival and continuation of a species on Earth.

Master Mind Map 🗺️



---

1. Do Organisms Create Exact Copies of Themselves? 🧬

Reproduction at its core involves making copies of the blueprints of body design.

- **DNA (Deoxyribonucleic Acid)** is the information source for making proteins in the cell.
- During cell division, DNA replication occurs. However, no biochemical reaction is absolutely reliable, meaning small **variations** occur during DNA copying.
- These variations are the basis for **Evolution**, helping a species survive changing environmental conditions.

QUICK TIP

Variation is beneficial for the survival of the *species* over time, even if it may not necessarily benefit an *individual* organism.

---

2. Modes of Asexual Reproduction

Asexual reproduction involves only a single parent and does not involve the fusion of gametes. The offspring are genetically identical to the parent.

Mode	Description	Examples
Fission	A single cell splits into two or more daughter cells. - *Binary Fission* (splits into two) - *Multiple Fission* (splits into many)	Amoeba, Leishmania Plasmodium (Malarial parasite)
Fragmentation	Multicellular organism breaks into pieces upon maturation. Each piece grows into a new individual.	Spirogyra (Algae)
Regeneration	The ability of a fully differentiated organism to regenerate the whole body from its body parts.	Hydra, Planaria
Budding	A small outgrowth (bud) develops on the parent body due to repeated cell division, detaches when mature, and becomes a new individual.	Hydra, Yeast

Spore Formation	Parent plant produces hundreds of microscopic reproductive units called spores. When the spore case bursts, spores spread and grow into new plants under favorable conditions.	Rhizopus (Bread mould)
Vegetative Propagation	New plants are obtained from the parts of old plants (stems, roots, leaves) without the help of reproductive organs.	Bryophyllum (leaves), Potato (stem), Rose (stem cutting)

---

3. Sexual Reproduction in Flowering Plants 🌺

The reproductive parts of angiosperms (flowering plants) are located in the **flower**.

Parts of a Flower

1. **Sepals**: Protect the flower in the bud stage.
2. **Petals**: Brightly colored to attract insects for pollination.
3. **Stamen (Male Part)**: Consists of an **Anther** (produces pollen grains) and a **Filament** (stalk).
4. **Carpel/Pistil (Female Part)**: Consists of **Stigma** (sticky top to catch pollen), **Style** (tube connecting stigma to ovary), and **Ovary** (contains ovules/egg cells).

IMPORTANT POINTS 📌

- **Unisexual flowers**: Contain either stamens OR carpels (e.g., Papaya, Watermelon).
- **Bisexual flowers**: Contain BOTH stamens AND carpels (e.g., Hibiscus, Mustard).

Pollination and Fertilization 🐝

- **Pollination**: The transfer of pollen grains from the anther to the stigma of a flower.
  - *Self-pollination*: Transfer within the same flower or same plant.
  - *Cross-pollination*: Transfer to a different plant of the same species.
- **Fertilization**: When a pollen grain lands on the stigma, it grows a pollen tube down the style to the ovary. The male germ-cell fuses with the female germ-cell (egg) inside the ovule. This fusion is called fertilization.
- After fertilization, the **zygote** divides to form an **embryo** within the ovule. The ovule develops a tough coat and becomes the **seed**. The ovary grows rapidly and ripens to form the **fruit**.

---

4. Reproduction in Human Beings 👪

Humans reproduce sexually. The reproductive organs become functional at **Puberty** (the period during adolescence when sexual maturity is reached).

A. Male Reproductive System 🚹

- **Testes**: Located outside the abdominal cavity in the scrotum (because sperm formation requires a lower temperature than normal body temperature). They produce **sperms** and the male hormone **Testosterone**.
- **Vas Deferens**: A tube that carries sperms from the testes.
- **Prostate Gland & Seminal Vesicles**: Add their secretions (fluid) to the sperms to provide nutrition and make transport easier. This fluid + sperms = **Semen**.
- **Penis**: Passes urine and semen out of the body.

B. Female Reproductive System 🚺

- **Ovaries**: Produce female egg cells (ova) and hormones (**Estrogen** and **Progesterone**). Every month, one egg matures and is released.
- **Fallopian Tube (Oviduct)**: Carries the egg from the ovary to the womb. **Fertilization of the egg by sperm occurs here.**
- **Uterus (Womb)**: A muscular bag where the fertilized egg (zygote) is implanted and develops into a baby.

- **Cervix & Vagina**: Forms the birth canal.

> ⚠️ **COMMON MISTAKES** ❌
>
> **What happens when the egg is NOT fertilized?** 🩸 Because the uterus prepares itself every month to receive a fertilized egg, its lining becomes thick and spongy. If fertilization does not occur, this lining is not needed any longer. It slowly breaks and comes out through the vagina as blood and mucous. This cycle takes place roughly every month and is known as **Menstruation**.

---

5. Reproductive Health & Birth Control 🛡️

Reproductive health broadly means total well-being in all aspects of reproduction (physical, emotional, behavioral, and social).

Contraception is the avoidance of pregnancy. It can be achieved by:

1. **Mechanical Barriers**: Condoms (prevent sperm from reaching the egg). Also protect against STDs (Sexually Transmitted Diseases).
2. **Chemical Methods**: Oral pills (change hormonal balance so eggs are not released).
3. **IUCD (Intrauterine Contraceptive Devices)**: Copper-T placed in the uterus to prevent pregnancy.
4. **Surgical Methods**:
   - *Vasectomy* in males (blocking vas deferens).
   - *Tubectomy* in females (blocking fallopian tubes).

Quick Revision Notes 📌

1. **DNA Replication** creates variations leading to evolution.
2. **Vegetative Propagation** is faster and helps plants that have lost the capacity to produce seeds (e.g., Banana, Rose) grow.
3. **Placenta**: A special disc-like tissue embedded in the uterine wall. It provides a large surface area for glucose and oxygen to pass from the mother's blood to the

developing embryo, and waste from embryo to mother.

4. **STDs**: Gonorrhoea, Syphilis (Bacterial); Warts, HIV/AIDS (Viral). Condoms are the only contraceptive method that helps prevent STDs.

Chapter Summary 🏁

Reproduction is vital for species survival. Asexual methods involve single parents with identical offspring and are common in simpler organisms. Sexual reproduction involves two parents, creates genetic variation, and happens in complex plants and animals. In flowers, pollination brings male pollen to female ovules. In humans, complex specialized organs facilitate internal fertilization and womb development. Finally, reproductive health awareness and contraception methods ensure healthy lives and controlled population growth.